RIG-SMOOTH MORPHISMS ARE STABLE UNDER COMPOSITION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is complete: the six-term sequence, the annihilation argument, the localization short exact sequence and the final criterion application are all written out

Editorial note. The label lemma-composition-rig-smooth-continuous names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the rig-smooth criterion via H^0 of the completed naively trivialized cotangent complex being locally finite projective and H^{-1} killed by a power of the ideal of definition
- the long exact cohomology sequence from the transitivity triangle for NL
- the annihilator lemmas for the two-term complex under base change
- localization and extensions of finite projective modules

Source reference. Original remark 0GBK: sources/stacks-chow/tags.txt, source lines 20375–20375. Prerequisite pointer invoked at proof line(s) 4398; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GAJ: sources/stacks-restricted/pinned-original-source.tex, source lines 843–843. Prerequisite pointer invoked at proof line(s) 4423; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 00S2: sources/stacks-restricted/pinned-original-source.tex, source lines 725–725. Prerequisite pointer invoked at proof line(s) 4409; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GCI: sources/stacks-restricted/pinned-original-source.tex, source lines 4214–4214. Prerequisite pointer invoked at proof line(s) 4404; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:restricted.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0G9H: sources/stacks-chow/tags.txt, source lines 20302–20302. Prerequisite pointer invoked at proof line(s) 4426; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0G9K: sources/stacks-chow/tags.txt, source lines 20305–20305. Prerequisite pointer invoked at proof line(s) 4432; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0G9G: sources/stacks-chow/tags.txt, source lines 20301–20301. Prerequisite pointer invoked at proof line(s) 4425; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *The property $P(\varphi) =$"φ is rig-smooth" on arrows of $WAdm^{Noeth}$ is stable under composition as defined in Formal Spaces, Remark 0GBK.*

Proof. We strongly urge the reader to find their own proof and not read the proof that follows. The statement makes sense by Lemma 0GCI. To see that it is true assume we have rig-smooth morphisms $A \to B$ and $B \to C$ in $WAdm^{Noeth}$. Then we can choose an ideal of definition $I \subset A$ such that the topology on C and B is the I-adic topology. By Lemma 00S2 we obtain an exact sequence

$$C \otimes_B H^0(NL^\wedge_{B/A}) \longrightarrow H^0(NL^\wedge_{C/A}) \longrightarrow H^0(NL^\wedge_{C/B}) \longrightarrow 0$$

$$H^{-1}(NL^\wedge_{B/A} \otimes_B C) \longrightarrow H^{-1}(NL^\wedge_{C/A}) \longrightarrow H^{-1}(NL^\wedge_{C/B})$$

Observe that $H^{-1}(NL^\wedge_{B/A} \otimes_B C)$ and $H^{-1}(NL^\wedge_{C/B})$ are annihilated by a power of I; this follows from Lemma 0GAJ part (2) combined with More on Algebra, Lemmas 0G9G and 0G9H (to deal with the base change by $B \to C$). Hence $H^{-1}(NL^\wedge_{C/A})$ is annihilated by a power of I. Next, by the characterization of rig-smooth algebras in Lemma 0GAJ part (2) which in turn refers to More on Algebra, Lemma 0G9K part (5) we can choose $f_1, \ldots, f_s \in IB$ and $g_1, \ldots, g_t \in IC$ such that $V(f_1, \ldots, f_s) = V(IB)$ and $V(g_1, \ldots, g_t) = V(IC)$ and such that $H^0(NL^\wedge_{B/A})_{f_i}$ is a finite projective B_{f_i}-module and $H^0(NL^\wedge_{C/B})_{g_j}$ is a finite projective C_{g_j}-module. Since the cohomologies in degree -1 vanish upon localization at $f_i g_j$ we get a short exact sequence

$$0 \to (C \otimes_B H^0(NL^\wedge_{B/A}))_{f_i g_j} \to H^0(NL^\wedge_{C/A})_{f_i g_j} \to H^0(NL^\wedge_{C/B})_{f_i g_j} \to 0$$

and we conclude that $H^0(NL^\wedge_{C/A})_{f_i g_j}$ is a finite projective $C_{f_i g_j}$-module as an extension of same. Thus by the criterion in Lemma 0GAJ part (2) and via that the criterion in More on Algebra, Lemma 0G9K part (4) we conclude that C is rig-smooth over (A, I). □